

# JOEL E. BOON

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## EDUCATION

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### PRINCETON UNIVERSITY

Princeton, New Jersey

#### B.S.E. - Mechanical and Aerospace Engineering

(Expected) May 2027

- Relevant Coursework: Thermodynamics; Fluid Mechanics; Fluid Dynamics; Engineering Design; Mechanical Design; Automatic Control Systems; Space Flight; Space Systems Design; Rockets and Airbreathing Propulsion Technology

## PROJECTS

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### VAMPS SPACE MISSION DESIGN

Princeton, New Jersey

#### Space Systems Design

January 2026 – May 2026

- Analyzed and designed GNC and ADC subsystems within a sub-team to fulfill a NASA style Phase A mission design
- Wrote subsystem requirements and coordinated with other teams to support an integrated mission design

### AIRCRAFT WING

Princeton, New Jersey

#### Engineering Design

March 2025 – May 2025

- Collaborated with a team to design, manufacture, and test a wing to hold an end load of 97 lbf (final failure at 300 lbf)
- Performed static and buckling analyses using FEA to optimize the number and spacing of bulkheads in the wing

### AUTONOMOUS SEARCH AND RESCUE ROBOT

Princeton, New Jersey

#### Mechanical Design

September 2025 – December 2025

- Designed and built a lifting arm mechanism to autonomously lift and translate a 20 lb robot
- Utilized PTC Creo (CAD) and finite element analysis (FEA), manufactured parts to meet subsystem requirements

### PINBALL MACHINE

Howard County, Maryland

#### HiTech Summer STEM Program

August 2025

- Led a small team to design, prototype, test, and manufacture a wooden pinball kit for engineering classes
- Met design requirements while reducing costs by 60% compared with the previous commercial kit

## EXPERIENCE

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### GRACE COMMUNITY CHURCH

Howard County, Maryland

#### Pastoral Intern

June 2026 – August 2026

- Prepared and delivered a sermon to middle schoolers, translating theological concepts into a clear, engaging message
- Researched biblical passages and wrote devotions, connecting theological themes to practical experiences

### HOWARD COUNTY LIBRARY SYSTEM

Howard County, Maryland

#### HiTech Summer STEM Program – Lead Instructor

May 2024 – August 2025

- Developed new curricula for week-long STEM classes for secondary school students, refined existing programs
- Worked with colleagues to construct two strongman machines and a pinball machine for the annual library carnival

### ANDERSEN CUSTOM KITCHENS

Catonsville, Maryland

#### Carpenter Assistant

June 2021 – August 2023 (Seasonal)

- Assisted in fabricating and assembling custom cabinets, applying lean principles to improve workflow efficiency

## LEADERSHIP

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### AEROPRINCETON

Princeton, New Jersey

#### Head Officer of RC Plane Division

February 2026 - Present

- Oversaw creation of the RC plane division of AeroPrinceton, created systems engineering framework for RC planes

### PRINCETON CHRISTIAN FELLOWSHIP

Princeton, New Jersey

#### President

February 2026 – Present

- Led weekly events for 50+ students, delegated leadership responsibilities, coordinated with university administration

## SKILLS/INTERESTS

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**Skills:** MATLAB, Simulink, STK, Java, Fusion 360, PTC Creo, Woodworking, Manufacturing, and LaTeX  
**Interests:** Woodworking, Princeton Club Flag Football, speed solving Rubik's cubes